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Thermodynamic modeling of the La-Te system aided by first-principles calculations

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ABSTRACT

A complete thermodynamic description of the La-Te binary system is developed by means of CALculation of PHAse Diagram (CALPHAD) method in combination with available experimental data in the literature and the present first-principles calculations based on density functional theory. The intermetallic phases with homogeneity ranges, $La_{3-x}Te_4$ and $LaTe_{2-x}$, are modeled using a two-sublattice $(La,Va)_3(Te)_4$ model and a three-sublattice $(La)_1(Te)_1(Te,Va)_1$ model based on their structure features, respectively. The intermetallic phases, LaTe and $LaTe_3$, are treated as stoichiometric compounds. The thermodynamic properties of the intermetallic compounds and their corresponding end members at finite temperatures are predicted using first-principles quasi-harmonic approach. The associate solution model is used to describe the short-range ordering behavior of the liquid phase. The calculated phase diagram agrees well with the available phase equilibrium data in the literature.

1. Introduction

The rare earth (RE) chalcogenide intermetallic compounds have constituted a subject of intensive studies because of their interesting physical properties and rich structural chemistry. Among the RE-chalcogenide systems, the lanthanum tellurides are of special interest for their appealing thermoelectric properties and thermal stability at high temperatures. For example, the n-type $La_{3-x}Te_4$ compound is identified as a promising candidate to replace conventional SiGe-based materials for thermoelectric power generation at high temperatures due to its higher thermoelectric efficiency above 800 °C [1,2]. The stoichiometric La_3Te_4 compound has also been well studied for its superconducting properties below 4 K [3,4]. In addition, the $LaTe_{2-x}$ and $LaTe_3$ compounds have attracted a great deal of attention for functional materials applications due to their unique electrical, optical and transport properties [5–7].

Understanding the phase equilibria as well as the thermochemistry of the La-Te system is essential for the synthesis and development of the lanthanum tellurides related materials. For example, the synthesis condition of the $La_{3-x}Te_4$ compound is restricted by the high vapor pressure of Te and the weak oxidation resistance of both components [8]. In addition, for the thermoelectric materials undergoing high temperature operation for long periods of time, issues related to sublimation must be considered [2], due to the high vapor pressure of Te, which results in loss of mass and subsequent phase decomposition, degrading the thermoelectric performance of the material consequently. Besides, the carrier concentration of the $La_{3-x}Te_4$ phase has been found to strongly depend on the crystal stoichiometry [9]. Experiments revealed that a metal-insulator transition exists when the value of x varies from 0 to 1/3 [9]. A comprehensive investigation on the thermodynamics of the La-Te system can provide valuable guidance and useful information for optimizing the synthesis conditions as well as the thermoelectric performance of the La-Te materials. Furthermore, for the thermoelectric devices operated at high temperatures, understanding the interfacial phase stability between the electrode and thermoelectric material is of great importance, since intensive chemical reactions may happen at the interface and damage the thermoelectric performance of the device. Ni is a candidate device component for use in thermoelectric devices due to its electrical and thermal characteristics [10]. In order to investigate the phase equilibria at the Ni/ $La_{3-x}Te_4$ interface, phase equilibria and thermodynamic properties of the constituent binary systems must be studied.

The CALculation of PHAse Diagrams (CALPHAD) method has been proven to be an invaluable method to model the thermodynamic

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properties of the individual phases and predict the equilibria between the phases in a multi-component materials system [11]. The CALPHAD method establishes thermodynamic models for all individual phases in the system of interest to describe their Gibbs energy functions, and optimizes the model parameters based on the computational and experimental data on thermochemistry and phase equilibria. The model parameters can also be used for extrapolations to high-order systems. However, the available experimental data on thermochemistry are often scarce for many materials systems, which usually leads to a large uncertainty of the modeling results. Alternatively, first-principles calculations based on density functional theory (DFT) have been demonstrated to be a reliable method of predicting missing thermochemical data for CALPHAD modeling when experimental data are lacking or have large uncertainties [11]. In the present work, first-principles calculations with the quasi-harmonic approach were performed to predict the thermodynamic properties of the stoichiometric compounds and the end-members of the phases with solubility in the La-Te system as a function of temperature. The predicted thermochemical data provide a means to model the system with CALPHAD method more accurately. A self-consistent thermodynamic description of the La-Te system is provided and will be used for the development of the thermodynamic database of the Ni-La-Te system to study the interface stability between Ni electrode and $La_{3-x}Te_4$-based thermoelectric materials in our future work.

2. Review of literature data

Some experimental investigations have been performed on the La-Te systems in terms of phase equilibrium and thermochemistry, notably those by Ramsey et al. [12], Eliseev et al. [13], and Delaire et al. [8]. Their experimental observations are described in detail in the remainder of this section. The crystallography information of the reported intermetallic phases is summarized in Table 1.

2.1. Phase equilibria

Ramsey et al. studied phase equilibria of the La-Te system in the whole composition range using thermal analysis and X-ray diffractography (XRD) [12]. The samples for phase boundary measurements were synthesized by annealing the solid mixtures with different compositions in two types of reaction tubes. The samples having Te composition greater than 66.7 at% were sealed in evacuated silica tubes at 873 K for 3 days to allow reactions, and then held at various temperatures for 2 weeks to reach equilibrium. For the samples with Te compositions lower than 66.7 at%, they were sealed in tantalum tubes and heated in vacuum at approximately 2273 K for several minutes, and then equilibrated at various temperatures for 2 weeks. After annealing, all the samples were quenched, and then XRD techniques were used to identify the phases in each sample. Four intermetallic phases were reported and considered to be stable, namely LaTe, $La_{3-x}Te_4$, $LaTe_{2-x}$ and $LaTe_3$ [12]. The LaTe and $LaTe_3$ phases are stoichiometric while the $La_{3-x}Te_4$ and $LaTe_{2-x}$ phases were reported to display solubility ranges, from 57.1 to 60 at% Te in $La_{3-x}Te_4$, and 62.7–66.7 at% Te in $LaTe_{2-x}$, respectively [12]. In addition, the LaTe phase was reported to melt congruently at 1993 ± 25 K [12]. The liquidus in the composition range between 0 and 50 at% Te were also determined by melting solid mixtures of pure La and LaTe in an inert atmosphere [12]. However, in their work, the eutectic point between bcc-La and LaTe was not determined and the invariant reaction temperature was postulated about 1168 K [12]. The $La_{3-x}Te_4$ phase was reported to melt congruently [12]. The melting temperatures of La_3Te_4 and $La_{8/3}Te_4$ (La_2Te_3) were measured to be 1788 ± 25 K and 1758 ± 25 K, respectively [12]. The $LaTe_{2-x}$ phase was proposed to melt above 1673 K incongruently but the corresponding invariant reaction was not investigated [12]. The $LaTe_3$ was found to undergo a peritectic decomposition at 1108 ± 15 K, and the peritectic point was unknown [12]. A eutectic reaction between the $LaTe_3$ and Te-rich hexagonal phase was found by a differential thermal analysis (DTA) at approximately 673 K; the eutectic composition was estimated to be more than 97 at% Te [12]. However, the experimental conditions of the DTA measurement were not reported in detail.

Eliseev et al. also investigated the homogeneity ranges of the La-Te intermetallic phases in the temperature range below 1273 K using XRD [13]. However, the experimental conditions (e.g. annealing temperature, annealing time and sample preparation) of their measurements were not described [13]. In addition to the four intermetallic phases observed by Ramsey et al. [12], two more phases, namely $LaTe_{1.7\pm x}$ and La_2Te_5, were reported by Eliseev et al. [13]. The $La_{3-x}Te_4$, $LaTe_{1.7\pm x}$ and $LaTe_{2-x}$ phases were reported to have the single-phase composition ranges of 57.2–60 at% Te, 61.0–64.0 at% Te and 64.2–66.5 at% Te, respectively [13]. In addition, Eliseev et al. indicated that the $LaTe_3$ phase also displayed a homogeneity range from 71.4 to 75.0 at% Te [13]. The DTA measurements were also performed in their work to determine the invariant reactions and liquidus boundary in the composition range of 0–50 at% Te and 70–100 at%Te. The samples used for DTA analysis were annealed in vacuum at 873–1273 K with the time varying from 50 to 100 h depending on the sample composition [13]. Their measurements yielded a eutectic reaction at 1168 ± 5 K (L ↔ bcc-La + LaTe), a peritectic reaction at 1086 ± 5 K (L + $LaTe_{2-x}$ ↔ La_2Te_5), a second peritectic reaction at 1054 ± 5 K (L + La_2Te_5 ↔ $LaTe_3$), and a second eutectic reaction at 728 ± 5 K (L ↔ $LaTe_3$ + Te) [13]. The compositions of these invariant reactions were not determined. The liquidus with the Te content higher than 75 at% was also determined [13]. The bcc-to-fcc and fcc-to-dhcp polymorphic transformation temperatures of La were measured as 1108 ± 5 K and 548 ± 5 K [13], whereas the accepted value for pure La are 1134 K and 550 K [14]. An additional heat absorption at 693 ± 5 K was observed for the samples with compositions between 0 and 50 at% Te by Eliseev et al. [13]. However, the reason of the existence of this additional heat absorption was not discussed by the authors [13].

Comparing the phase equilibria data from Eliseev et al. [13] with those from Ramsey et al. [12], there are following discrepancies:

(1) The $LaTe_{1.7\pm x}$ and La_2Te_5 phases reported by Eliseev et al. [13] are not found by Ramsey et al. [12]. At the composition of 61.9 at% Te, which is in the single-phase region of the $LaTe_{1.7\pm x}$ phase claimed by Eliseev et al. [13], Ramsey et al. [12] annealed the samples at five different temperatures between 600 and 1300 K as long as two weeks and found co-existence of the $La_{3-x}Te_4$ and $LaTe_{2-x}$ phases. In fact, the XRD pattern of the $LaTe_{1.7\pm x}$ phase obtained by Eliseev et al. [13] is very close to that of the $LaTe_{2-x}$ phase. Therefore, it is possible that the $LaTe_{1.7\pm x}$ phase reported by Eliseev et al. is actually a non-equilibrium state of the $LaTe_{2-x}$ phase. In addition, Ramsey et al. showed that the stoichiometry of La_2Te_5 is actually in a two-phase region between the $LaTe_{2-x}$ and the $LaTe_3$ phases [12].
(2) The solubility range of the $LaTe_{2-x}$ phase is reported as 64.2–66.5 at % Te by Eliseev et al. [13], while the value reported by Ramsey et al. is about 62.7–66.7 at% Te [12].
(3) The DTA measurements of Eliseev et al. yielded a peritectic decomposition of the $LaTe_3$ phase at 1054 ± 5 K [13]. However, the decomposition temperature reported by Ramsey et al. is

Table 1
Crystallographic data of the intermetallic phases in the La-Te binary system [9] together with details of the sublattice models for CALPHAD modeling.

Phases	Pearson symbol	Space group	Prototype	Composition at% Te	Sublattice model
LaTe	cF8	$Fm\bar{3}m$	NaCl	50.0	$(La)_1(Te)_1$
$La_{3-x}Te_4$	cI28	$I\bar{4}3d$	Th_3P_4	57.1	$(La, Va)_3(Te)_4$
$LaTe_{2-x}$	mP24	P1c1	$LaTe_2$	66.7	$(La)_1(Te)_1(Te, Va)_1$
$LaTe_3$	oS16	*Cmcm*	Mn_3As	75.0	$(La)_1(Te)_3$

1108 ± 15 K [12]. In addition, the $LaTe_3$ phase was reported as stoichiometric by Ramsey et al. [12]. Contrary to the results from Ramsey et al., Eliseev et al. indicated that the $LaTe_3$ phase has a solubility range from 71.4 to 75.0 at% Te [13].

(4) The temperature of the eutectic reaction between liquid, $LaTe_3$ and hexagonal Te was determined as 728 ± 5 K by Eliseev et al., but 673 K by Ramsey et al. [12].

For the sake of consistency, the evaluation of the modeling parameters is mainly based on the phase equilibria data from Ramsey et al. [12] in the present work. This is because the reliability of the intermetallic phase solubility data from Eliseev et al. [13] cannot be judged since they did not give the experimental conditions, such as the annealing temperature and duration time. In addition, their DTA measurements seem to be questionable for two reasons: (i) the temperature of the bcc-to-fcc polymorphic transformation of pure La was measured to be 26 K below the commonly accepted value [14]. (ii) The additional heat absorption at 693 ± 5 K implies that reaction might happen between the samples and impurities during heating process.

2.2. Thermochemical data

The thermochemistry measurements of the La-Te system are rather scarce and only performed for the $La_{3-x}Te_4$ phase by Delaire et al. [8]. They measured the heat capacity of the $La_{3-x}Te_4$ phase from 1.8 to 300 K using a Quantum Design physical property measurement system (PPMS), and from 400 to 1200 K using differential scanning calorimeter (DSC) at a nominal composition of 57.1 at% Te.

3. CALPHAD modeling

In the present work, the LaTe and $LaTe_3$ phase are treated as stoichiometric compounds based on the experimental data from Ramsey et al. [12], and their Gibbs energy above room temperature is expressed as follow,

$$G^{La_xTe_y} - H^{SER} = a + bT + cT\ln T + dT^2 + eT^{-1} + fT^3 \quad (1)$$

where a, b, c, d, e and f are the model parameters which are determinable from the Gibbs energy calculated from the quasi-harmonic first-principles methods. The details of first-principles calculations are presented in the following section. H^{SER} denotes the stable element reference (SER) state that refers to the enthalpies of dhcp-La and hexagonal-Te at 298.15 K and 1 bar. It should be noted extrapolating Eq. (1) to high temperatures could result in an unrealistically high value of the heat capacity, which may artificially make the modeled phase becoming stable again (above ~ 5000 K). Therefore, in the present work, the heat capacity curves of the stoichiometric phases, and as well as the end-members, are cut off at a temperature that is about 100 K above its melting temperature [15–17]. Above the cutoff temperature, a constant C_P value is assumed due to unavailability of data.

Sublattice model was used for the modeling of the $La_{3-x}Te_4$ and $LaTe_{2-x}$ phases with homogeneity ranges. The $La_{3-x}Te_4$ phase was modeled with a sublattice formula $(La,Va)_3(Te)_4$ according to its crystal structural feature [12]. The sublattice $(La,Va)_3$ means that all the La sites are equivalent, and with possible vacancy occupations. The Gibbs energy of the $La_{3-x}Te_4$ phase in per mole of formula is described as,

$$G_m^{La_{3-x}Te_4} = y_{La}\,{}^0G_{La:Te} + y_{Va}\,{}^0G_{Va:Te} + \frac{1}{7}RT(3y_{La}\ln y_{La} + 3y_{Va}\ln y_{Va}) + {}^{xs}G \quad (2)$$

where y_{La} and y_{Va} denote the site fractions of La and vacancy on the first sublattice, respectively, ${}^0G_{X:Te}$ presents the Gibbs energy of the two end-members of the $La_{3-x}Te_4$ phase, i.e. $(La)_3(Te)_4$ and $(Va)_3(Te)$, and ${}^{xs}G$ captures any excess energy associated with mixing of La and vacancy on the first sublattice. A three-sublattice model, $(La)_1(Te)_1(Te,Va)_1$ was used for the $LaTe_{2-x}$ phase based on its crystal structural feature. The last sublattice accounts for the fact that the vacancies have a strong tendency to occupy one type of Te sites [18]. The Gibbs energy function of the $LaTe_{2-x}$ phase is expressed as,

$$G_m^{LaTe_{2-x}} = y_{Te}G_{La:Te:Te} + y_{Va}G_{La:Te:Va} + \frac{1}{3}RT(y_{Te}\ln y_{Te} + y_{Va}\ln y_{Va}) + {}^{xs}G \quad (3)$$

where y_{Te} and y_{Va} denote the site fractions of Te and vacancy on the third sublattice, respectively, and ${}^{xs}G$ represents the excess Gibbs energy. In the present work, the Gibbs energies of the end-members in Eqs. (2) and (3) were described using Eq. (1) to fit the first-principles results. The excess Gibbs energy term in Eqs. (2) and (3) is described functionally by the Redlich–Kister polynomial [19],

$${}^{xs}G = y_iy_j\sum_{k=0}{}^kL_{i,j}(y_i - y_j)^k \quad (4)$$

where y_i and y_j denote the sublattice site fraction of species i and j (i,j = La, Te, Va), respectively, and ${}^kL_{i,j}$ is kth interaction parameter between the species, which is expressed as,

$${}^kL_{i,j} = {}^kA + {}^kBT \quad (5)$$

where ${}^kA$ and ${}^kB$ are the parameters to be evaluated in the model.

The Gibbs energies of the solid solution phases (G_m^{ϕ}) are described by the disordered substitutional solution model as given by,

$$G_m^{\varphi} = \Sigma_i x_i\,{}^0G_i^{\varphi} + RT\sum_i x_i\ln x_i + {}^{xs}G_{\varphi} \quad (6)$$

where R is the gas constant, x_i is the mole fraction of i = La and Te, ${}^0G_i^{\varphi}$ denotes the Gibbs energy of pure La and Te in the structure φ, which are taken from the SGTE-PURE5 database if corresponding data are available [14]. In addition, since the Gibbs energy descriptions of the pure Te in dhcp, fcc and bcc structures as well as the pure La in hexagonal-Te structure are not available in the SGTE-PURE5 database [14], first-principles calculations were performed to investigate the relative lattice stabilities. For example, the Gibbs energy of dhcp-Te is described as G (dhcp, Te) = GHSERTE + $\Delta E_{dchp\text{-}hex}$, where $\Delta E_{dchp\text{-}hex}$ is the difference of the 0 K ground state energy between dhcp-Te and hexagonal-Te from first-principles calculations, and the GHSERTE is Gibbs energy function of Te at the standard reference state from the PURE5 database [14]. The lattice stabilities of other structures (i.e., fcc-Te, bcc-Te and hexagonal-La) are calculated in the same manner. The calculation results are listed in Table 2. The excess Gibbs energy ${}^{xs}G_{\varphi}$ takes the same forms as Eqs. (4) and (5).

The thermodynamic properties of the liquid phase were modeled using the associate solution mode with LaTe as the associate, since the LaTe phase has the highest melting temperature and an asymmetric shape of the liquidus around. Accordingly, the Gibbs energy of the liquid phase is of form,

$$G_m^{Liq}(T, y_i) = \sum_i y_i\,{}^0G_i^{Liq}(T) + RT\sum_i y_i\ln y_i + {}^{xs}G_L \quad (7)$$

where i presents any of the three species, La, Te and LaTe, and y_i is the mole fraction of specie i in the liquid phase, and ${}^0G_{La}^{Liq}$ and ${}^0G_{Te}^{Liq}$ represent the Gibbs energy of pure La and Te liquid, respectively, which are taken from the SGTE compilation [14]. The Gibbs energy of the LaTe specie, ${}^0G_{LaTe}^{Liq}$, is described as,

$${}^0G_{LaTe}^{Liq} = \frac{1}{2}{}^0G_{La}^{Liq} + \frac{1}{2}{}^0G_{Te}^{Liq} + {}^{LaTe}A + {}^{LaTe}BT \quad (8)$$

where ${}^{LaTe}A$ and ${}^{LaTe}B$ are the parameters evaluated during modeling. The excess Gibbs energy ${}^{xs}G_L$ is also described functionally in the form of Eqs. (4) and (5) with the mole fractions of the three species.

All the model parameters are evaluated based on the phase equilibrium data from Ramsey et al. [12] and the first-principles thermochemical data from the present work using the PARROT module of the Thermo-Calc software. The complete and self-consistent thermodynamic parameters of the La-Te system are listed in Table 2.

Table 2
Thermodynamic parameters of the La-Te binary system.

Phase	Parameters	Values
Liquid (La,Te)	G^{Liq}_{LaTe}	$G^{Liq}_{La} + G^{Liq}_{Te} - 247,000 + 1.57T$
	${}^{0}L^{Liq}_{La,LaTe}$	$-4,830 + 2.24T$
	${}^{1}L^{Liq}_{La,LaTe}$	$6,320 - 3.41T$
	${}^{0}L^{Liq}_{LaTe,Te}$	$-110,000 - 29T$
	${}^{1}L^{Liq}_{LaTe,Te}$	$-47,000 - 8T$
dhcp phase (La,Te)	G^{dhcp}_{Te}	${}^{0}G^{hex}_{Te} + 12648$
fcc phase (La,Te)	G^{fcc}_{Te}	${}^{0}G^{hex}_{Te} + 14588$
bcc phase (La,Te)	G^{bcc}_{Te}	${}^{0}G^{hex}_{Te} + 29763$
hexagonal phase (La,Te)	G^{hex}_{La}	${}^{0}G^{bcc}_{La} + 2826$
LaTe phase	$G^{LaTe}_{La:Te}$	$298 < T < 2000$ $-327,677.372 + 314.008506T - 61.6840482T \ln T + 9.59003586 \times 10^{-3}T^2 + 292,391.813T^{-1} - 2.96156443 \times 10^{-6}T^3$
		$2000 < T$ $-383,502.516 + 596.900298T - 94.25526T \ln T$
$La_{3-x}Te_4$ phase $(La,Va)_3(Te)_4$	$G^{La_{3-x}Te_4}_{La:Te}$	$298 < T < 1900$ $-1,147,513.98 + 976.319026T - 195.144672T \ln T + 4.71669321 \times 10^{-3}T^2 + 462,098.679T^{-1} - 7.23028893 \times 10^{-6}T^3$
		$1900 < T$ $-1,327,884.09 + 2,099.35125T - 333.573289T \ln T$
	$G^{La_{3-x}Te_4}_{Va:Te}$	$298 < T < 1900$ $+108,667.195 + 472.199547T - 101.218451T \ln T - 6.15235432 \times 10^{-3}T^2 + 72,093.2955T^{-1} - 1.6803761 \times 10^{-6}T^3$
		$1900 < T$ $+40,506.1724 + 941.337248T - 160.954404T \ln T$
Interaction Parameters	${}^{0}L^{La_{3-x}Te_4}_{La,Va:Te}$	$-414,158 + 99.75T$
	${}^{1}L^{La_{3-x}Te_4}_{La,Va:Te}$	$-246,127-35.76T$
$LaTe_{2-x}$ phase	$G^{LaTe_{2-x}}_{La:Te:Te}$	$298 < T < 1800$ $-432,401.415 + 394.010787T - 80.8109987T \ln T + 3.66542342 \times 10^{-3}T^2 + 184,668.639T^{-1} - 1.82076277 \times 10^{-6}T^3$
$La(Te)_1(Te,Va)_1$		$1800 < T$ $-462,589.826 + 577.076722T-102.897111T \ln T$
	$G^{LaTe_{2-x}}_{La:Te:Va}$	$298 < T < 1800$ $-233,572.063 + 292.094764T - 59.686377T \ln T + 5.76656934 \times 10^{-3}T^2 + 232,086.95T^{-1} - 2.70175331 \times 10^{-6}T^3$
		$1800 < T$ $-277,399.13 + 555.150113T - 91.305548T \ln T$
Interaction Parameter	${}^{0}L^{LaTe_{2-x}}_{La:Te:Te,Va}$	$-217,039 + 17.14T$
	${}^{1}L^{LaTe_{2-x}}_{La:Te:Te,Va}$	$+50,183$
$LaTe_3$ phase	$G^{LaTe_3}_{La:Te}$	$-448,694.857 + 478.015491T - 100.128259T \ln T - 4.11649854 \times 10^{-3}T^2 + 74,667.4593T^{-1} - 1.02772391 \times 10^{-6}T^3$

Table 3
Calculated lattice parameters of pure elements and intermetallic phases in the La-Te system from first-principles calculations in comparison with available experimental data. The details of calculations settings of each phases, including k-point mesh for structure relaxation, and supercell size and k-point mesh for phonon calculations, are also listed.

Phases	Reference	Lattice parameter (Å)			k-Point mesh electron	Supercell size	k-Point mesh phonon
		a	b	c			
dhcp-La	Present work	3.762		12.075	16 × 16 × 5	2 × 2 × 2	3 × 3 × 3
	Expt. [34]	3.756		12.150			
hexagonal-Te	Present work	4.498		5.967	13 × 13 × 9	2 × 2 × 2	3 × 3 × 3
	Expt [33]	4.440		5.920			
LaTe	Present work	6.486			13 × 13 × 14	4 × 4 × 4	3 × 3 × 3
	Expt [12]	6.436					
$(La)_3(Te)_4$	Present work	9.688			11 × 11 × 11	2 × 2 × 2	3 × 3 × 3
	Expt [12]	9.628					
$(La)_1(Te)_1(Te)_1$	Present work	9.206	9.081	9.080	10 × 10 × 10	2 × 2 × 2	3 × 3 × 3
	Expt [18]	9.190	9.107	9.070			
$LaTe_3$	Present work	4.428		27.838	6 × 6 × 1	2 × 2 × 2	3 × 3 × 1
	Expt [12]	4.410		26.100			

4. First-principles methodology

First-principles calculations based on density functional theory were performed to predict the thermochemical properties of the stoichiometric compounds and end-members at finite temperatures. Based on the quasi-harmonic approach, the Helmholtz energy of a condensed phase as a function of temperature (T) and volume (V) consists of three parts [20],

$$F(V, T) = E_0(V) + F_{ele}(V, T) + F_{vib}(V, T) \quad (9)$$

The first term, $E_0(V)$, is the 0 K static energy without the contribution of the zero-point vibration. This term can be determined by a

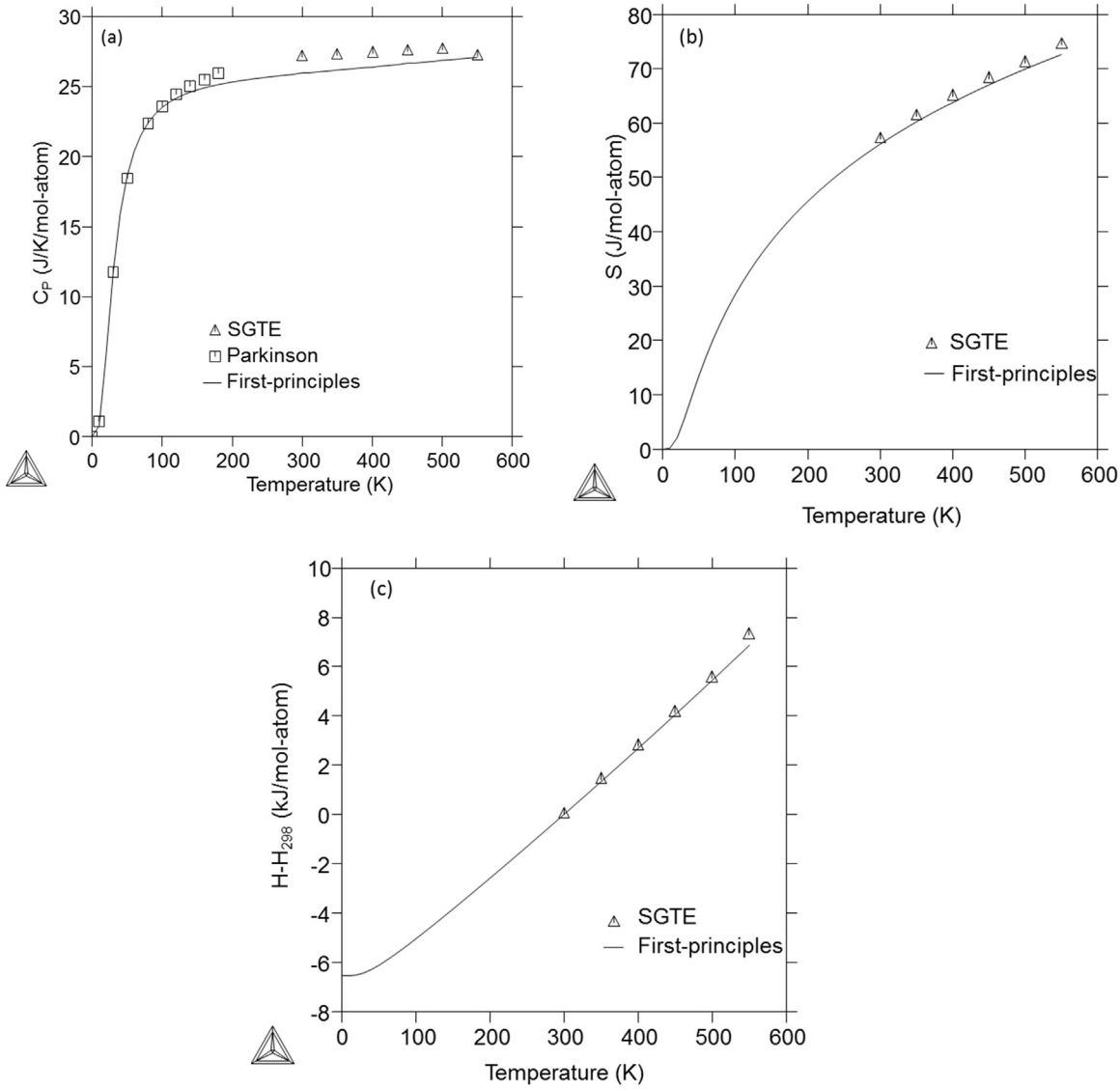

Fig. 1. Finite-temperature thermodynamic properties of dhcp-La: (a) heat capacity (b) entropy and (c) enthalpy as a function of temperature predicted from the present first-principles quasi-harmonic approach along with available experimental data [35] (□) and the SGTE/PURE 5 database [14] (△).

fitting of the first-principles energy vs. volume data points using a four-parameter Birch-Murnaghan equation of state (EOS) [21,22],

$$E(V) = a + bV^{-2/3} + cV^{-4/3} + dV^{-2} \tag{10}$$

where a, b, c and d are fitting parameters. The minimum value of the fitting is the 0 K static energy at the equilibrium state, and the corresponding volume is the equilibrium volume (V_0) of the phase at 0 K. In the present work, the data points for the EOS fitting were obtained by performing first-principle calculations at seven fixed volumes. A volume relaxation of the input supercell was first performed to obtain a start volume, V′. Then, the seven different volumes for the EOS fitting were chosen based on V′, namely 0.871 V′, 0.913 V′, 0.956 V′, 1.000 V′, 1.046 V′, 1.093 V′, 1.141 V′, which are corresponding to an incremental change of 1.5% in the length of the three lattice parameters of the simulation supercell.

The second part in Eq. (9) accounts for the thermal electronic contribution to the Helmholtz energy, which can be written as [20,21],

$$F_{ele} = E_{ele} - TS_{ele} \tag{11}$$

where E_{ele} and S_{ele} are the energy and entropy of thermal electron excitations, respectively. The entropy term can be calculated via integrating the electronic density of state (DOS) along with the distribution of the states over the energies of the electrons:

$$S_{ele}(V, T) = -k_B \int n(\varepsilon, V)[\ln f(\varepsilon, T) + (1 - f(\varepsilon, T))\ln(1 - f(\varepsilon, T))]\mathrm{d}\varepsilon \tag{12}$$

where $n(\varepsilon, V)$ is the electronic density of state at energy ε, and f is the Fermi-Dirac distribution since electrons are fermions and given by:

$$f(\varepsilon, T) = \left[\exp\left(\frac{\varepsilon - \mu}{k_B T}\right) + 1\right]^{-1} \tag{13}$$

where μ is the chemical potential of electrons. The energy term can be calculated by:

$$E_{ele}(V, T) = \int_{-\infty}^{\infty} n(\varepsilon, V) f(\varepsilon, T)\varepsilon \mathrm{d}\varepsilon - \int_{-\infty}^{\varepsilon_f} n(\varepsilon, V)\varepsilon \mathrm{d}\varepsilon \tag{14}$$

where ε_f is Fermi energy. In the present work, the electronic DOS is obtained from first-principles calculations.

The last part in Eq. (9), $F_{vib}(V,T)$, represents the contribution of lattice vibrations to the Helmholtz energy. Under the quasi-harmonic approximation, this part of contribution can be calculated based on either the phonon DOS [20] or the Debye–Grüneisen model [21]. The phonon method is generally accepted to be the more accurate approach

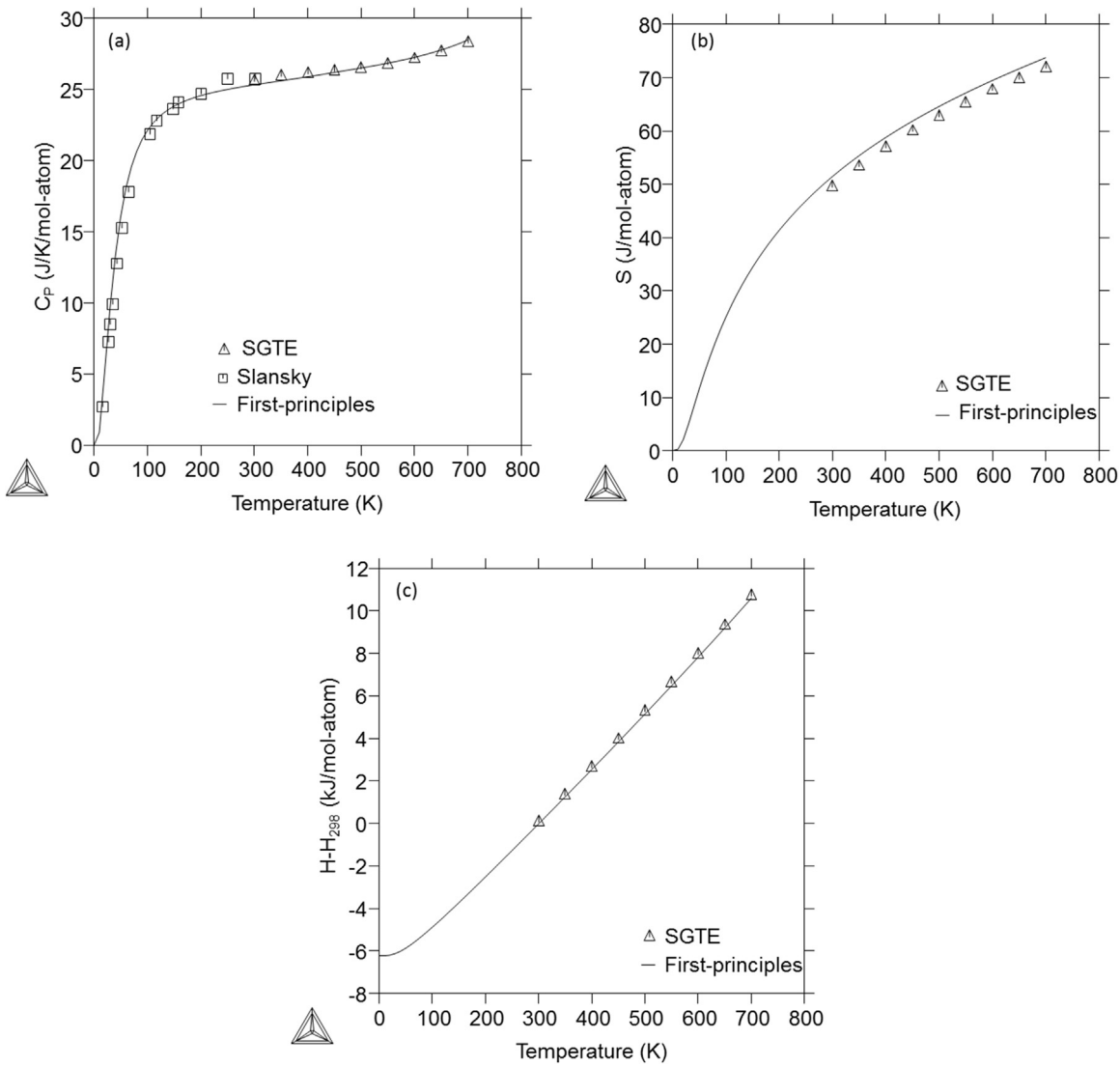

Fig. 2. Finite-temperature thermodynamic properties of hexagonal-Te: (a) heat capacity (b) entropy and (c) enthalpy as a function of temperature predicted from the present first-principles quasi-harmonic approach along with available experimental data [36] (□) and the SGTE/PURE 5 database [14] (△).

for predicting the finite-temperature thermodynamic properties. However, this method cannot be applied for the dynamically unstable structures, such as the unstable end-members in CALPHAD modeling. Therefore, in the present work, the vibrational Helmholtz energy of the stable stoichiometric phases and end-members were calculated using the phonon method, and the Debye–Grüneisen model was employed for the calculations of the dynamically unstable end-members.

Based on the partition function of lattice vibration, the quasi-harmonic phonon approach describes the vibrational contribution to Helmholtz energy, $F_{vib}(V,T)$, as follows [20,21],

$$F_{vib}(V,\ T) = k_B T \int_0^{\infty} \ln\left[2\sinh\frac{\hbar\omega}{2k_B T}\right] g(\omega,\ V)d\omega \tag{15}$$

where $\hbar$ is the reduced Plank constant, ω the phonon frequency, and $g(\omega,\ V)$ denotes the phonon DOS at the frequency ω and volume V. The phonon DOSs of each phase of interest are calculated for the same seven volumes as those for EOS fitting. In the present work, the phonon calculations were performed for the dhcp-La, hexagonal-Te, LaTe and $LaTe_3$ phases, and two stable end-members, namely $(La)_3(Te)_4$ and $(La)_1(Te)_1(Te)_1$. In the quasi-harmonic Debye–Grüneisen model, $F_{vib}(V,T)$ term is expressed as [21],

$$F_{vib}(V,\ T) = \frac{9}{8}k_B\Theta_D + k_B T\left\{3\ln\left[1-\exp\left(-\frac{\Theta_D}{T}\right)\right] - D\left(\frac{\Theta_D}{T}\right)\right\} \tag{16}$$

where k_B is the Boltzmann constant and $D(\Theta_D/T)$ the Debye function. The Debye temperature Θ_D can be calculated by,

$$\Theta_D = s(6\pi^2)^{\frac{1}{3}}\frac{\hbar}{k_B}V_0^{\frac{1}{6}}(B_0/M)^{\frac{1}{2}}(V/V_0)^{\gamma} \tag{17}$$

where M is the atomic mass, V_0 is the ground state volume and B_0 is the corresponding bulk modulus. γ is the Grüneisen constant, which is defined as $\gamma = [(1+B_0')/2 - x]$. B_0' is the first derivative of the bulk modulus with respect to pressure. The values of V_0, B_0, and B_0' are obtained from the EOS fitting. The value of x is 2/3 when temperature is higher than Debye temperature and 1 is for low temperature case [21,23]. Here, x = 2/3 is used because the high temperature data will have a greater impact on the Gibbs energy functions for the CALPHAD modeling. s is a scaling factor for Debye temperature. Moruzzi et al. gave a value of 0.617 for 14 nonmagnetic cubic metals [24]. This value may also be applicable for materials which do not fall into this category, as demonstrated in our previous modeling work [25,26] and can also be calculated from elastic constants when needed [27]. In the present work, the Helmholtz energy of the two unstable end-members, $(Va)_3(Te)_4$ and (La)1(Te)1(Va)1, were obtained using the quasi-

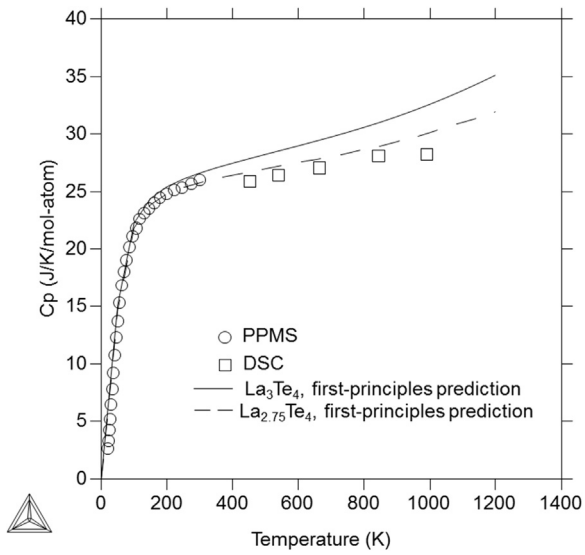

Fig. 3. Calculated heat capacity of the $(La)_3(Te)_4$ endmember (solid line) from the present first-principles calculations in comparison with the results of the $La_{2.75}Te_4$ structure (dash line) in which one La-vacancy was introduced in the $(La)_3(Te)_4$ supercell, and the experimental measurements at a nominate composition of La_3Te_4 [8] (PPMS (○) and DSC (□)).

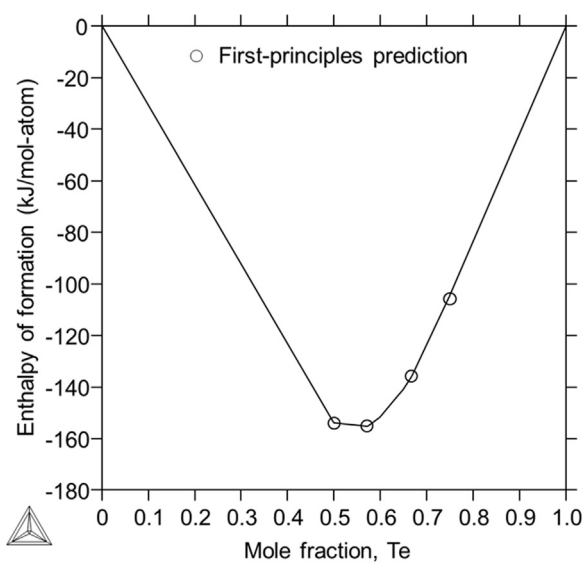

Fig. 4. Enthalpies of formation of the La-Te system at 298 K from the present CALPHAD modeling (line), in comparison with the predictions from the first-principles calculations (○).

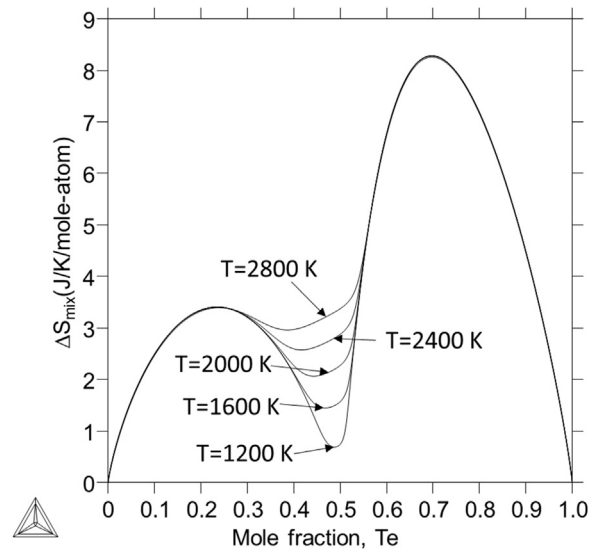

Fig. 5. Calculated entropy of mixing in the liquid at 2800 K, 2400 K, 2000 K, 1600 K and 1200 K from the present CALPHAD modeling.

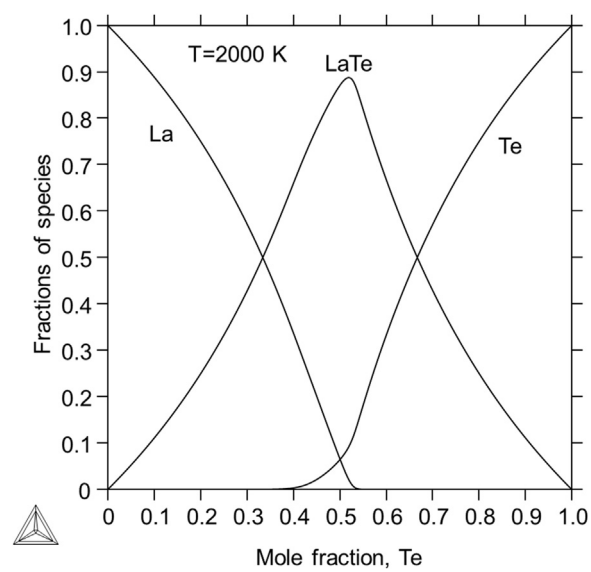

Fig. 6. mol fractions of species in the liquid phase as a function of mole fraction of Te at 2000 K, calculated from the present CALPHAD modeling.

harmonic Debye–Grüneisen model.

The Vienna Ab-initio Simulation Package (VASP) was used as the computational tool for the first-principles calculations in the present work. The projector augmented wave method (PAW) [28,29] and the exchange-correlation functional depicted by the generalized gradient approximation by Perdew, Burke, and Ernzerhof (PBE) [30] were employed. The valance electronic states were $5s^25p^6d^16s^2$ and $5s^2p^4$ for La and Te, respectively, as recommended by the VASP code. The energy convergence criterion of the electronic self-consistency was set as 10^{-6} eV/atom for all the calculations. The atomic structures were relaxed by implementing the Methfessel-Paxton method in order to calculate the forces acting on the atoms [31]. The Brillouin-zone integration of static calculations is performed by using a tetrahedron method with Blöchl corrections. The phonon calculations were carried out using the supercell method. The Force constants, i.e. the Hessian matrix, were calculated directly using the VASP code. The phonon density of states (DOS) were predicted using the YPHON code based on a parameter-free, mixed-space approach [32]. The k-points grid and simulation supercell size used for structural relaxations and phonon calculations for each phase are provided in Table 3.

5. Results and discussion

To provide a benchmark for the first-principles methodology used in the present work, the calculated theoretical lattice parameters of the pure elements, the LaTe and $LaTe_3$ phases, and the stable end-members of the $La_{3-x}Te_4$ and $LaTe_{2-x}$ phases are compared with available experimental data [12,18,33,34]. As shown in Table 3, the calculated lattice parameters agree well with the experimental data within 1.5% error, except for the c-axis of the $LaTe_3$ phase. It is found that the theoretical calculations overestimate the c-axis of the $LaTe_3$ phase about 7% when compare to the experimental data [12]. This may be because the van der Waals interactions between the layered structures in the $LaTe_3$ phase were not considered by the present first-principles

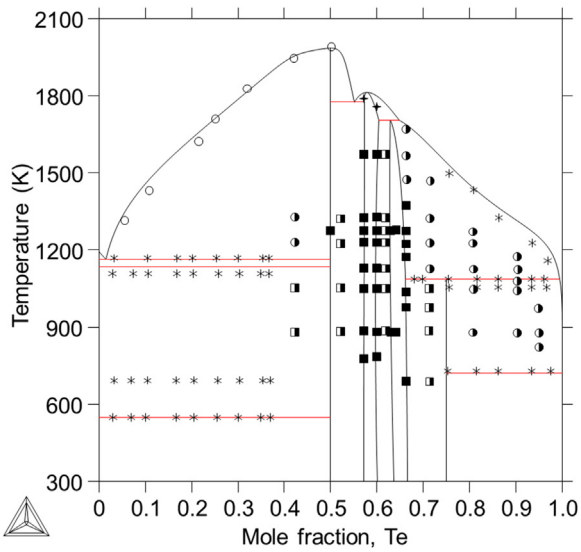

Fig. 7. Calculated La-Te phase diagram with the experimental data from Ramsey et. al. [12] (○ -liquidus, ✦ - melting, ◑ - solid-liquid two-phase region, ◨ - solid-solid two-phase region, ■ - single phase region) and Eliseev et. al. [13] (✳).

calculations.

To further validate the results from the quasi-harmonic approach, the calculated thermodynamic properties of the pure dhcp-La and hexagonal-Te at finite temperatures are compared with the data from the SGTE-PURE5 database [14] and other available experimental data [35]. Fig. 1 shows the heat capacity, entropy and enthalpy of pure dhcp-La from 0 K to 550 K, respectively. The calculated heat capacity is in good agreement with experimental values below 200 K. At higher temperatures, one can see that the present calculations underestimate the heat capacity slightly. However, this discrepancy is considered inconsequential for the following CALPHAD modeling since the calculated entropy and enthalpy show excellent agreement with the SGTE-PURE5 data, as shown in Fig. 1b and c. The calculated heat capacity, entropy and enthalpy for pure hexagonal-Te are plotted in Fig. 2, respectively, showing good agreement with the SGTE-PURE5 data [14] and experimental data from Slansky et. al. [36].

The predicted heat capacity of the $(La)_3(Te)_4$ endmember from DFT calculations is also compared to the experimental heat capacity of the $La_{3-x}Te_4$ phase, which is measured at a nominal composition of La_3Te_4 [8]. As shown in Fig. 3, the predicted heat capacity agrees well with the experimental data below 300 K. However, the discrepancies between the calculation results and experimental data become larger at higher temperatures. According to the experimental investigations from Ramsey et. al. [12], vacancies can occupy the La sites in the $La_{3-x}Te_4$ phase to result in a solubility range from the composition of La_3Te_4 to La_2Te_3. Therefore, for the experimental sample with a nominal composition of La_3Te_4, vacancy defects may be introduced during either the synthesis process via mechanical alloying or the process of heat capacity measurement at high temperatures, resulting in a La-deficient composition from its nominated value. Correspondingly, the first-principles calculations were also performed for the $La_{3-x}Te_4$ structure with vacancies occupying the La sites. One vacancy defect was introduced on a La atomic site in the supercell which is comprised of 12 La atoms and 16 Te atoms before vacancy substitution, giving a composition of $La_{2.75}Te_4$. As shown by the dash line in Fig. 3, the calculated heat capacity of $La_{2.75}Te_4$ is much lower than that of the La_3Te_4 structure which has no vacancy on La site (solid line), and in fairly good agreement with the experimental data. This implies that the actual composition of the experimental sample may not be the nominal value of La_3Te_4. Additionally, the discrepancies between theoretical and experimental data at higher temperatures may also correspond to the possible uncertainties of the DSC measurements, since it is well known that the DSC measurements at higher temperatures are relatively difficult and prone to errors. Therefore, in the present work, instead of using the experimental data, the theoretical heat capacity of the La_3Te_4 structure from DFT calculations is used for the evaluation of the modeling parameters of the $(La)_3(Te)_4$ endmember.

The evaluation of the CALPHAD modeling parameters of the intermetallic phases in the La-Te system was based on the thermochemical properties predicted by first-principles calculations. Fig. 4 shows the enthalpy of formation of the La-Te system at 298 K from the current CALPHAD modeling, along with the predictions from the first-principles calculations. As shown, the results of CALPHAD modeling are in excellent agreement with the DFT predictions. In addition, more comprehensive comparisons between the first-principles calculations and CALPHAD modeling results of heat capacity, entropy and enthalpy for the La-Te compounds and corresponding end-members are shown in supplementary material (Fig. S1). It should be noted that the entropy parameters of some compounds were further modified to reproduce the experimental phase equilibria accurately, which results in small discrepancies between the modeling results and the first-principles calculations (Fig. S1).

The phase diagram of the La-Te system exhibits several features indicating possible existence of local short-range ordering in the liquid phase, such as the steep liquidus and the stable intermetallic phases having much higher melting temperatures than their constituent elements. Therefore, in the present work, the liquid phase was modeled using the associate model with LaTe as the associate. This model usually gives a better physical description of the short-range ordering behavior

Table 4
Summary of the invariant reactions in the La-Te system.

Reaction type	Reaction	Composition (at%Te)			T/K	Method	Reference
Eutectic	Liquid ↔ bcc_A2 + LaTe		~0.00	50.00	1168	Experimental[a]	[13]
		1.53	~0.00	50.00	1164	Calculated	This work
Congruent	Liquid ↔ LaTe		50.00		1993 ± 25	Experimental[b]	[12]
			50.00		1984	Calculated	This work
Eutectic	Liquid ↔ LaTe + $La_{3-x}Te_4$	55.11	50.00	57.22	1775	Calculated	This work
Congruent	Liquid ↔ $La_{3-x}Te_4$		57.91		1813	Calculated	This work
Peritectic	Liquid + $La_{3-x}Te_4$ ↔ $LaTe_{2-x}$	64.81	60.42	62.88	1705	Calculated	This work
Peritectic	Liquid + $LaTe_{2-x}$ ↔ $LaTe_3$				1108 ± 25	Experimental[b]	[12]
		99.49	66.14	75.00	1086	Calculated	This work
Eutectic	Liquid ↔ $LaTe_3$ + hexagonal_A8	~97.00	75.00	~100.00	673	Experimental[c]	[12]
		~100.00	75.00		728	Experimental[a]	[13]
		~100.00	75.00	~100.00	723	Calculated	This work

[a] Differential scanning calorimetry measurements from Eliseev et. al. [13].
[b] Experimental measurements from Ramsey et. at. [12]. The melting temperature was determined using optical pyrometer.
[c] Differential scanning calorimetry measurements from Ramsey et. at. [12].

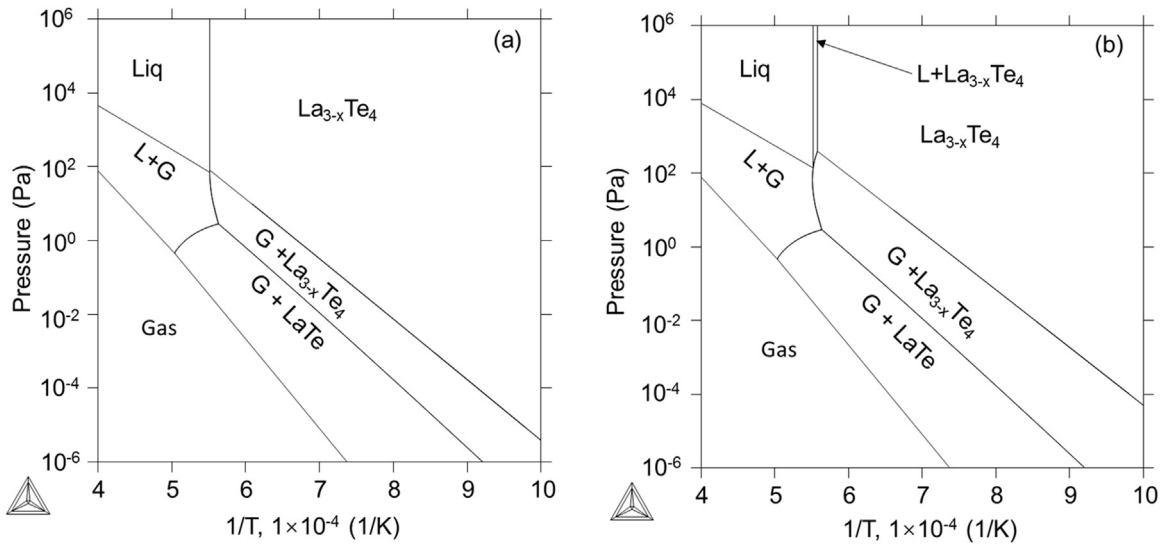

Fig. 8. Calculated pressure-temperature phase diagrams at two compositions: (a) 58.0 at% Te and (b) 58.8 at% Te.

in the liquid phase than the sub-regular solution model [16,37], thus yielding more reliable parameters for the La-Te system. The short-range ordering behavior can be further examined by plotting the entropy of mixing of the liquid phase. As shown in Fig. 5, there is a deep valley in the plot, which demonstrates that the short-range ordering in the liquid significantly decreases the entropy of mixing at the composition of 50 at % Te, and thus results in steep liquidus around this composition. Fig. 6 shows the calculated species fraction in liquid as a function of Te concentration at 2000 K. It is noted that around the mole fraction x (Te) = 0.5 the dominant species in the liquid phase is the LaTe associate, which also indicates the strong ordering interactions between the atoms in the liquid phase around this composition.

Fig. 7 shows the calculated temperature-composition phase diagram of the La-Te system along with the related experimental data [12,13]. The calculated liquidus agrees well with the experimental measurements from Ramsey et. al. [12] and Eliseev et. al. [13]. The homogeneity ranges of $La_{3-x}Te_4$ and $LaTe_{2-x}$ also match the experimental results excellently. The melting points of $La_{3-x}Te_4$ were reported to be 1788 ± 25 K and 1758 ± 25 K for the compositions of 57.1 and 60.0 at% Te [12], respectively, which are also well reproduced using the present thermodynamic model. The calculated results for the invariant reactions are compared with available experimentally data, shown in Table 4, depicting good agreement within the experimental uncertainties.

For the thermoelectric materials undergoing high temperature operation for long periods, sublimation may happen at low-pressure conditions. Shown in Fig. 8 is the calculated pressure-temperature phase diagram to investigate the sublimation behavior of the $La_{3-x}Te_4$ phase. The thermodynamic expression of the gas phase is taken from the SGTE Substances database [38]. The calculations are performed at two single-phase compositions of the $La_{3-x}Te_4$ phase, namely 58.0 and 58.8 at% Te. As seen in Fig. 8, the $La_{3-x}Te_4$ phase starts to sublimate around the pressure of 100 Pa and 1700 K. As the total pressure of the system decreases, the sublimation occurs at lower temperatures. If the pressure of the system is below the phase region of G + $La_{3-x}Te_4$ + LaTe, i.e. the line between the G + $La_{3-x}Te_4$ and G + LaTe two-phase regions, the $La_{3-x}Te_4$ phase will decompose to LaTe and gas phases. The deficiency of La in the $La_{3-x}Te_4$ phase results in a higher sublimation pressure due to its decreased stability, as shown in Fig. 8b.

6. Conclusion

Experimental phase equilibrium and thermodynamic data of the La-Te system available in the literature were critically reviewed. First-principles calculations were performed to predict the thermodynamic properties of the intermetallic phases in the La-Te system using the quasi-harmonic approach. The first-principles predictions provide robust input thermochemical data for the thermodynamic modeling of solid phases. The thermodynamic behaviors of the liquid phase are well described by the associate solution model. A complete set of self-consistent thermodynamic parameters were obtained based on the available experimental data and the first-principles predictions via the CALPHAD method. Comprehensive comparisons between the calculated and measured thermodynamic quantities indicate that the existing experimental information in the La-Te system can be satisfactorily accounted for by the present thermodynamic description. The obtained thermodynamic database is used to predict the stability of the thermoelectric phase $La_{3-x}Te_4$ and is the building block for the La-Te-Ni ternary database in an effort to understand interface stability between Ni and thermoelectric materials in our future work.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.calphad.2018.03.003.

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